

PES1UG24CS613

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SECTION K

Requirements & Priority Log

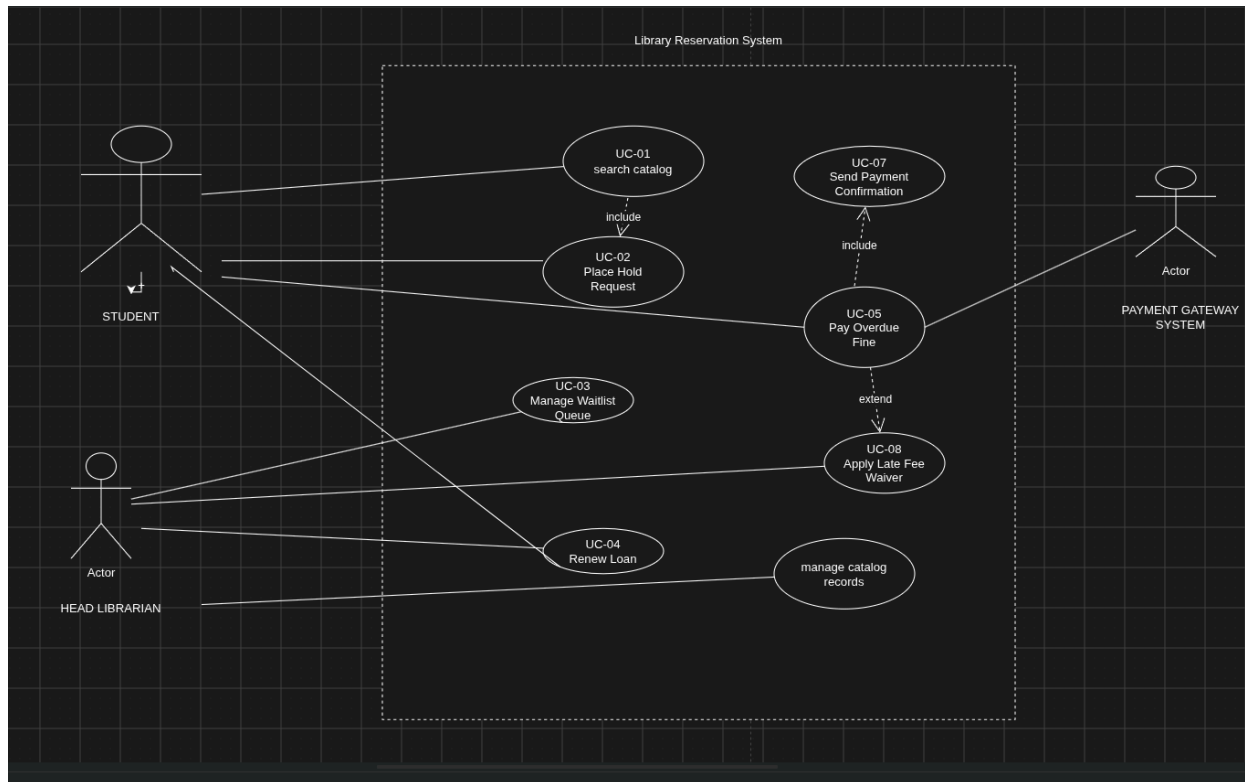
Digital Campus Library Reservation Gateway

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Student Member to search the ISBN catalog and place a hold request on a currently loaned book, entering a FIFO waitlist.	Hi... ▾	Pass: member is assigned the correct position in the waitlist queue. Fail: a member already holding the maximum allowed books is able to place an additional reservation.	Core reservation functionality; ensures equitable, order-based access to high-demand titles.
FR-002	Functional	The system shall automatically notify the next member in the FIFO waitlist (via email/SMS) within 5 minutes of a held book being checked back in as available.	Hi... ▾	Pass: notification timestamp is within 5 minutes of check-in and sent to the correct next member. Fail: notification is sent to the wrong member or	Prevents wasted hold-shelf capacity and gives members timely pickup windows.

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				delayed beyond 5 minutes.	
FR-003	Functional	The system shall allow a Student Member to renew an active loan up to 2 times, provided no other member is waiting on that title.	M... ▾	Pass: renewal extends the due date by 14 days and updates the loan record. Fail: renewal is granted despite an active hold on the title, or after 2 prior renewals.	Supports continued access to materials while protecting fairness for waitlisted members.
FR-004	Functional	The system shall allow a Student Member to pay an overdue fine through the integrated Payment Gateway and update the member's fine balance upon confirmed transaction.	Hi... ▾	Pass: fine balance is reduced by the correct amount and a receipt is generated only after gateway confirmation. Fail: balance updates without gateway confirmation, or a transaction is applied twice.	Enables self-service fine settlement, reducing front-desk workload and payment errors.
FR-005	Functional	The system shall allow the Head Librarian to view, and if necessary cancel or expire, any active hold in a title's waitlist queue (e.g., if a notified member fails to collect the book within 48 hours).	M... ▾	Pass: an expired or cancelled hold is removed from the queue and the next member is automatically promoted and notified. Fail: an expired hold remains active or blocks the next member's notification.	Keeps the FIFO queue accurate and prevents holds from being indefinitely stuck.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional (Performance & Security)	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms.	Hi... ▾	Pass: benchmarking tests confirm the target latency and security standards under simulated peak load.	Ensures a responsive search experience during peak enrollment/exam periods.
NFR-002	Nonfunctional (Security)	The system shall encrypt all Payment Gateway transactions and fine-related data in transit using TLS 1.2 or higher, and shall lock a member account for 15 minutes after 5 consecutive failed login attempts.	Hi... ▾	Pass: a security scan confirms TLS 1.2+ on all payment endpoints and the account locks automatically on the 5th failed login. Fail: any payment request is transmitted unencrypted, or the account is not locked after 5 failures.	Protects sensitive financial and personal data and mitigates brute-force login attacks.

UML DIAGRAM



Use-Case Flow Specification

Project: Digital Campus Library Reservation Gateway

Use Case: UC-02 — Place Hold Request

Primary Actor:

- Student Member (initiates search and hold request)

Secondary Actors:

- Head Librarian (queue oversight, notification trigger)

Includes:

- UC-01 Search Catalog («include» — hold requests always begin with a catalog search)

Preconditions:

- The Student Member is authenticated and logged into the library portal.

Postconditions:

- Success: The member holds a confirmed queue position for the requested title, and the system's waitlist record is updated.
- Failure: No hold is created and the member's reservation count remains unchanged.

Main Success Scenario:

1. Student Member logs into the library portal and selects "Search Catalog." System displays a search form (title, author, or ISBN).
2. Member enters search criteria; system executes UC-01 Search Catalog and returns matching results with availability status.
3. Member selects a title currently marked "on loan" and clicks "Place Hold."
4. System verifies the member's eligibility (active status, hold count < 3).
5. System adds the member to the title's FIFO waitlist and calculates the member's queue position.
6. System displays a confirmation screen showing the queue position and estimated availability window.
7. System logs the hold request with a timestamp for librarian reporting.
8. Use case ends successfully.

Alternate Flow — 2a. No Matching Title Found:

1. Search returns zero results, or the title is already available on the shelf (no hold needed).
2. System suggests the member borrow the title directly instead of placing a hold, and the use case ends.

Alternate Flow — 4a. Member Not Eligible:

1. System detects the member already holds the maximum of 3 active reservations.
2. System displays a "Reservation Limit Reached" message and does not add the member to the waitlist.
3. Member may cancel an existing hold and retry, or end the use case.

